Complete source-to-vertex transport measurements

All 23,000 actual pair transports pass both public P and Q image checks against the frozen destination fixtures. The selected graph has 46 vertices, 90 directed maps and five levels. Each of 100 paired targets is transported five times to every vertex.

The original first-discovery paths have zero to 8 maps. Each map costs 174 reference field multiplications per P/Q pair, including inverse exponentiation, and two inversions. Multiplication counts already include squaring and inversion exponentiation: do not add them twice. The empty source path has zero field arithmetic; its nonzero elapsed time measures loop/clock overhead.

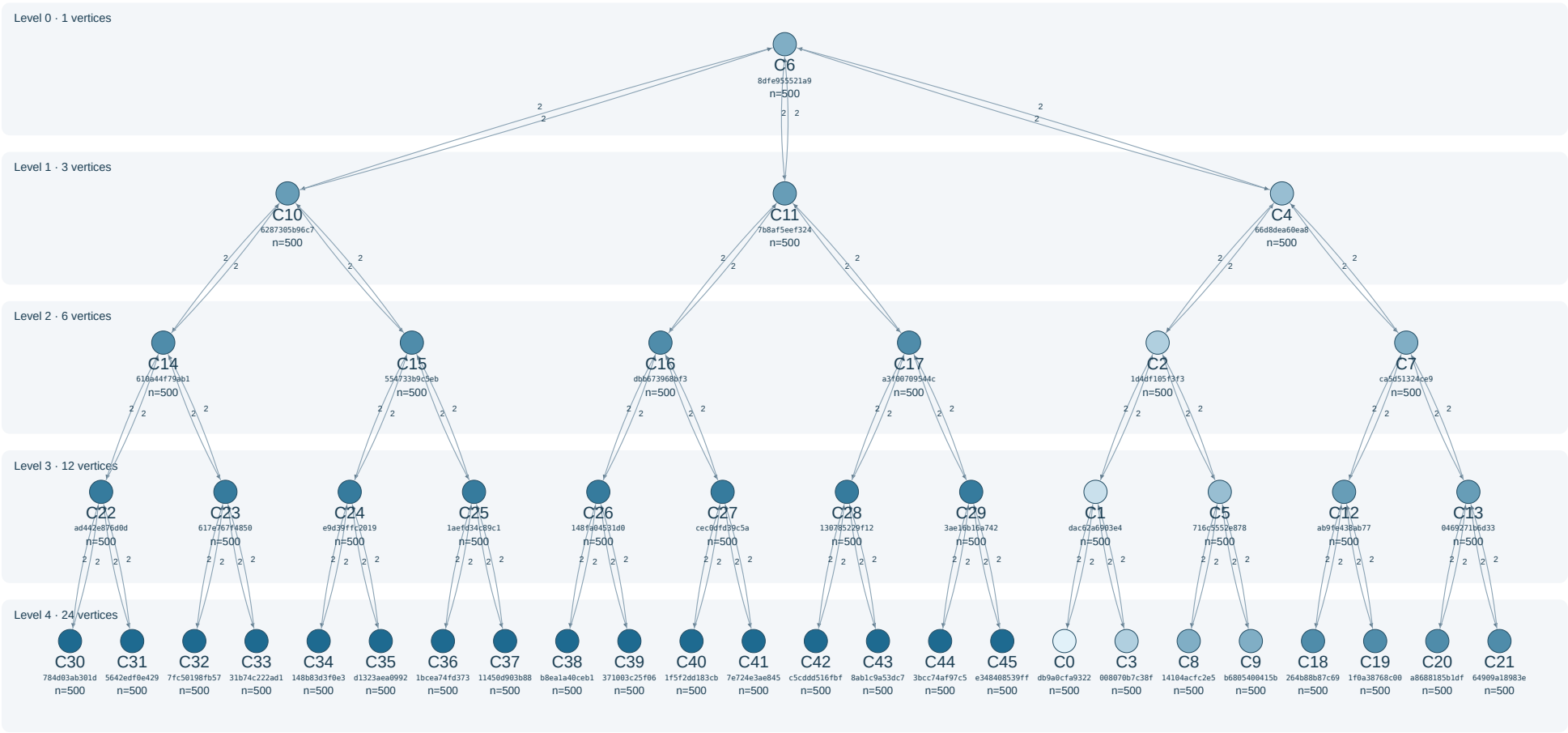
Every P and Q has order 18014494963342201 before and after transport. Longer routes cost more mapping operations without implying a harder ECDLP. Five repetitions are not five independent random scalars. Path preparation, parsing, membership/order checks and image validation are outside the measured interval. Reference maps and native Montgomery solver arithmetic have different counter definitions.

These are CPU-only concurrent L0 measurements. The elapsed range is 0.372–5705.426 microseconds across all observations; scheduler outliers remain in the min/mean/max tables. No transport mean is added to an unrelated solve and described as an observed end-to-end measurement. Solver arithmetic replays and random-corpus IC recovery remain separate work.

Sources: TRANSPORT_PROTOCOL.md, TRANSPORT_FROZEN.json, source/transport_measure.cpp and source/phase_report.mjs. Exact paths and per-curve/per-problem tables accompany this report. All original raw observations remain local.

Full-path P/Q transport: field multiplications

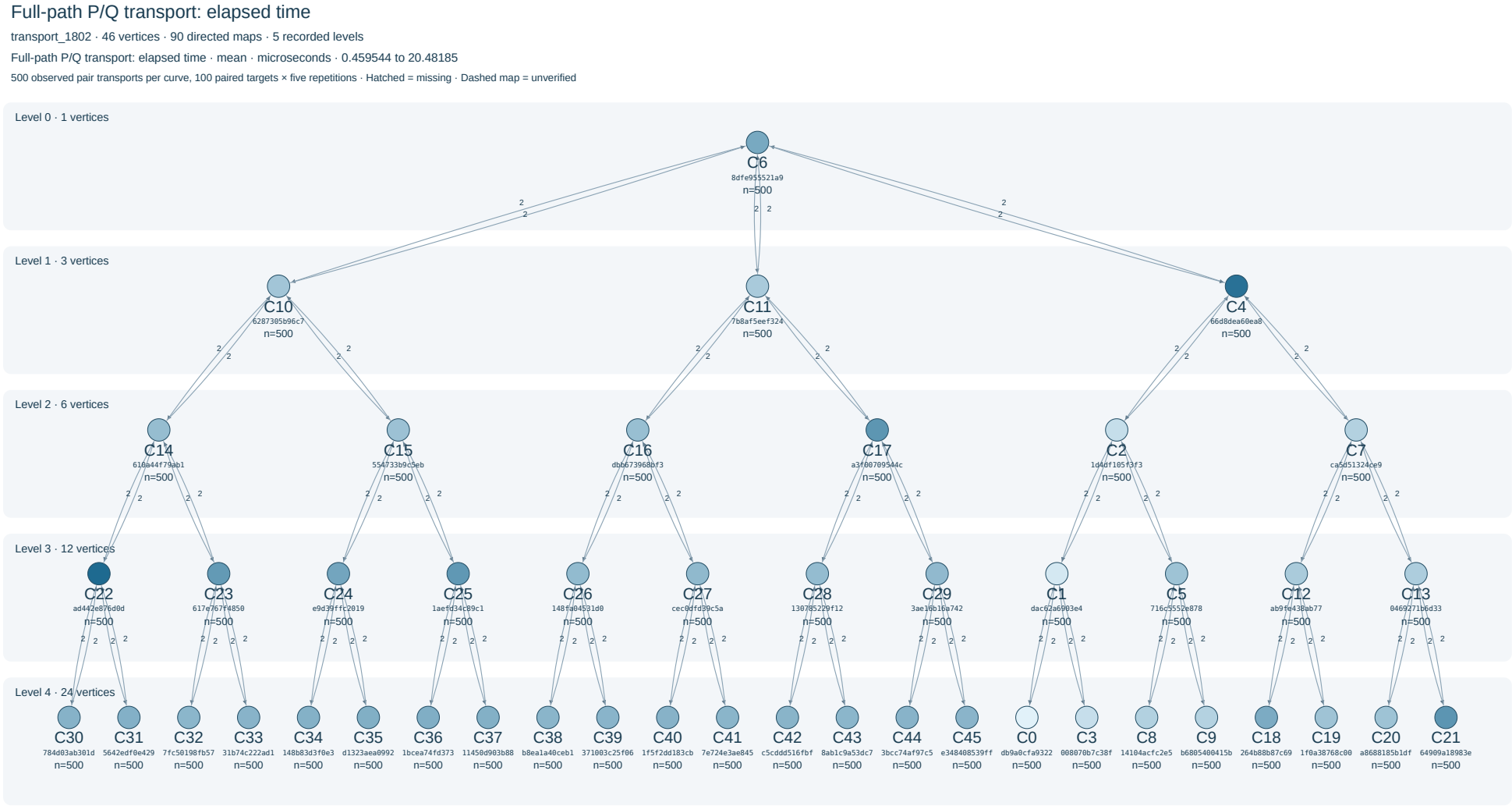
Actual reference operations; path preparation and image validation excluded. Unit: F_p multiplications including inverse exponentiation. Exact statistics and curve identities: transport_operations/graph_data.json.



Full exact identities, map evidence and measurements: graph_data.json / interactive inspector.

Full-path P/Q transport: elapsed time

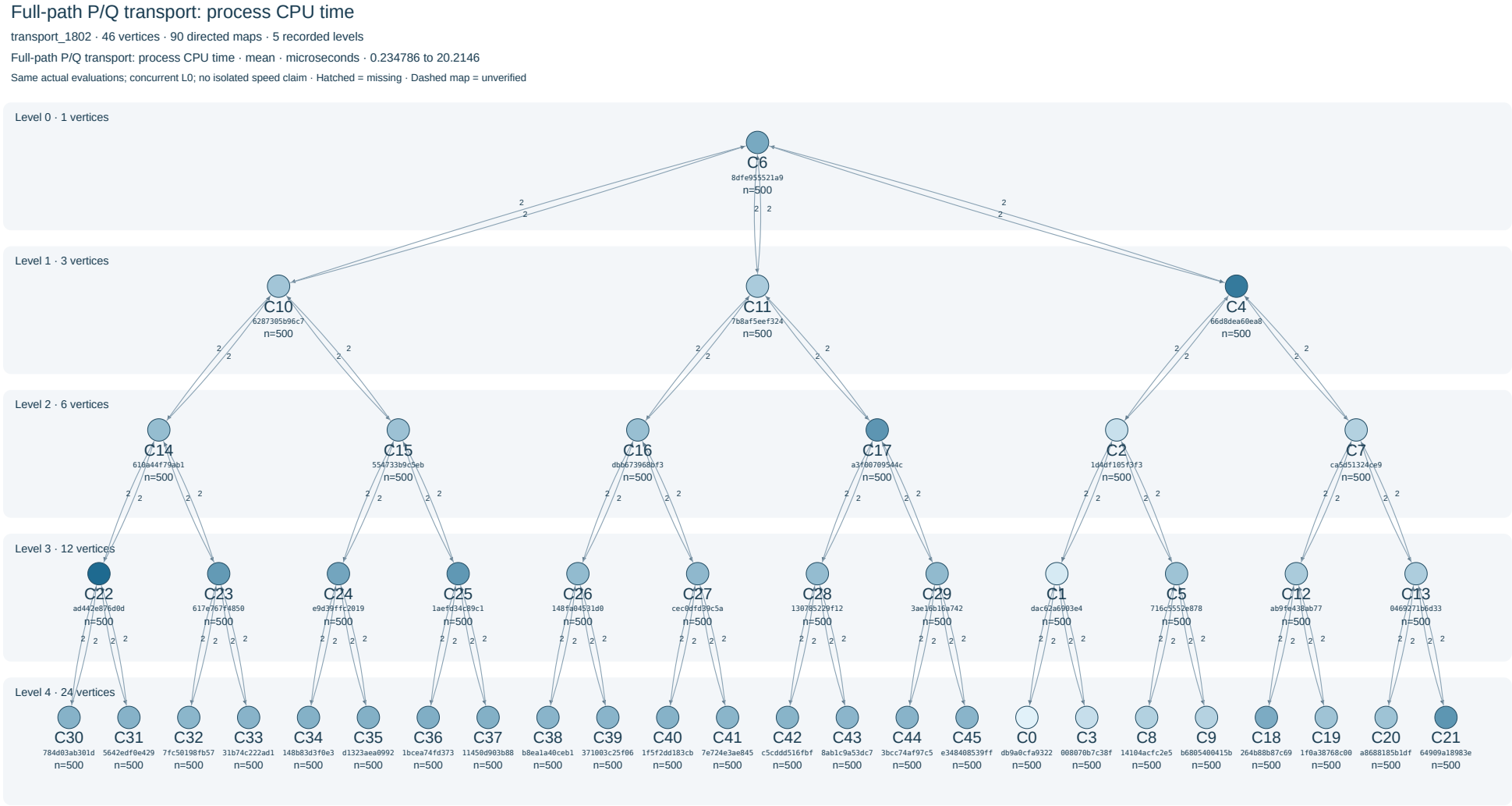
500 observed pair transports per curve, 100 paired targets × five repetitions. Unit: microseconds. Exact statistics and curve identities: transport_wall/graph_data.json.



Full exact identities, map evidence and measurements: graph_data.json / interactive inspector.

Full-path P/Q transport: process CPU time

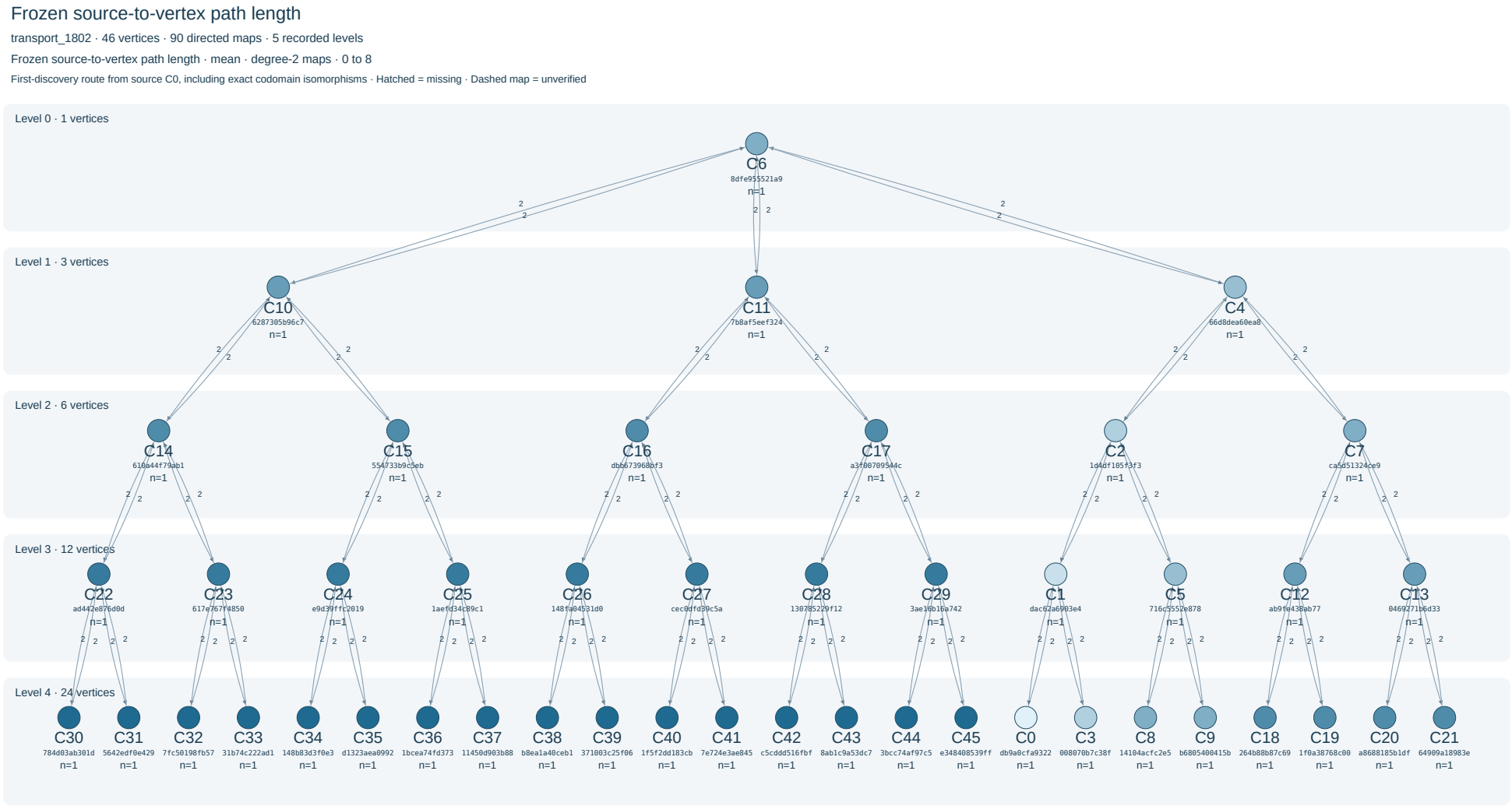
Same actual evaluations; concurrent L0; no isolated speed claim. Unit: microseconds. Exact statistics and curve identities: transport_cpu/graph_data.json.



Full exact identities, map evidence and measurements: graph_data.json / interactive inspector.

Frozen source-to-vertex path length

First-discovery route from source C0, including exact codomain isomorphisms. Unit: degree-2 maps. Exact statistics and curve identities: transport_steps/graph_data.json.



Full exact identities, map evidence and measurements: graph_data.json / interactive inspector.